A Closed-Form Equivariant Navigation Reflex under Bounded Observation Delay

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Abstract

We study pairwise disc robots under sampled double-integrator dynamics, bounded deterministic sensing error, and bounded asynchronous observation age, with no communication and no online optimizer. The controller is an explicit equivariant map: goal PD plus a radial push and a left-chiral tangent, with declared age and error tubes subtracted from clearance. On a named transit class C, Lipschitz delay-on safety and finite-horizon ranking certificates stand through horizon $k{=}16$, with a derivation caveat on the Lipschitz activation gate. The same observation convention that feeds a high-order control-barrier quadratic program makes that filter optimistic: delayed neighbor minus current own inflates separation, and a collinear constant-velocity identity plus two frozen contacts record true barrier $h{<}0$ against naive $h{>}1$. A four-cell necessity matrix on class C then separates controller from tubes: the closed-form law with tubes is the only cell that stands; wrap-none HOCBF and the tube-less reflex each admit a linearized continuum collision box, and HOCBF with the same tubes is infeasible or non-pass-through on named subsets. The paper is scoped. It is not a deadlock-free theorem, not a novelty freeze, and not a claim that HOCBF is dominated.

1 Introduction

The charter cell is deliberately small. Agents are planar discs with acceleration-limited double-integrator plants. Sensing error is a deterministic bound, not a Gaussian chance envelope. Observation age is bounded, asynchronous, and *not* a communication delay: each robot reconstructs the neighbor from a stale measurement of that neighbor against its own current state. There is no message passing and no quadratic or linear program at runtime. Safety is a sampled-data certificate on a named set, not an informal simulation score.

Three contributions occupy that cell and no larger one.

1. **Observation identity.** Under the charter wrap, naive reconstructed separation on a collinear constant-velocity head-on pair equals true separation plus half the product of closing speed and age. The resulting barrier is strictly optimistic whenever the pair is closing. Two frozen delay-corpus contacts exhibit true $h{<}0$ while naive $h{>}1$.

2. **Standing Lipschitz certificates.** On named class C as an initial set, delay-on Lipschitz safety of the closed-form tube reflex stands on the ranking box through $k{=}16$ and on the containing envelope E through $k{=}8$, together with finite-horizon ranking through $k{=}16$ and compact delay-on N-agent covers. The activation gate used to derive the Lipschitz constant is false; the constant itself is not falsified. Those covers stand with that caveat.

3. **Necessity matrix.** Crossing the closed-form law against an HOCBF-QP filter, and crossing declared tubes against their absence, yields a $2{\times}2$ of hashed continuum objects. Tubes are the causal safety ingredient. Only the closed form remains live under them.

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Non-claims. This draft does not assert deadlock-free progress (Gate 6), does not assert $\Phi \leq 0.35$, does not assert that "HOCBF is dominated," and does not assert coupled inertial arrival. Continuum covers that used a one-shot Taylor / zero-order-hold ball remainder (`f14f60c3`, `92c4f8cc`, `d76e27bd`, `8f86dfe9`, `d92f72bf`, `78ddd1e7`, `d835f618`, `d0f621b1`, `85abd8d4`) are withdrawn pending re-certification; physical vertex rollouts in those artifacts may be cited as vertices, never as continuum covers (`403f9d55...`; synthesis `dc644f3b...`). Gate 7 is a miss list, not a novelty freeze. Quantitative sentences below cite a hashed JSON prefix (Appendix A); the source of truth is the restricted synthesis (`dc644f3b...`).

2 Related work

Related work is a conservative miss list against the charter conjunction: double-integrator discs, bounded deterministic error, bounded asynchronous *observation* delay, no communication, no online optimizer, and a sampled-data certificate. Occupying a nearby cell is not occupying this one. A live arXiv refresh on 28 August 2026 updated the frontier matrix;[1] PaperCept was not recrawled: the 20 August 2026 pass (7278 titles, 0 delay$\wedge$CBF after title-and-abstract audit) remains the recorded proceedings miss (`8c95e4e7...`), and the IROS 2026 program URL still returns HTTP 403 (`5e3cf043...`). It is not a theorem that the cell is empty.

2.1 Velocity obstacles and Voronoi

Reciprocal velocity obstacles (ORCA/RVO2) compute a low-dimensional linear program over velocity halfplanes and need no communication on the pairwise disc plant [1]. They do not jointly certify acceleration limits, bounded deterministic sensing error, and observation age, and they are not claimed here under asynchronous delay. Buffered uncertainty-aware Voronoi cells (B-UAVC) insert a Gaussian chance buffer into a Voronoi projection [2]. A chance σ is not an adversarial tube, and delay is not part of that certificate.

2.2 Control barrier quadratic programs

Ames et al. encode a control barrier as an affine constraint and filter a nominal through a quadratic program [3]. High-order barriers extend the construction past relative degree one [4, 5]. Safety barrier certificates for heterogeneous multi-robot discs are the natural pairwise specialization [6, 7]. Every construction in this family is an online optimizer. Under the charter observation wrap the reconstructed barrier is optimistic (Section 5), so a feasible QP is not a delay certificate on this plant.

2.3 Delay and sampled-data barriers

Control barrier *functionals* treat state delay in an infinite-dimensional state and still synthesize a pointwise QP [8]. Input-delay CBF and integral CBF predictors compensate actuation lag, again with a QP; the March 2026 integral-CBF predictor is an adaptive-cruise example, not a communication-free disc reflex [9, 10, 11]. Capelli et al. keep connectivity with a CBF under communication delay by a tuned heuristic on inter-robot distance, not a declared observation tube [12]. Ballotta and Talak learn a distributed CBF with a graph-neural predictor of neighbor state under *communication* delay and age-of-information [13]. Zero-order CBFs for sampled-data systems and matrix CBFs for sampled-data multi-agent invariance remain filters (optimization or semidefinite programs), not a closed-form cell [14, 15, 16, 17]. A dedicated delay-CBF query set on 28 August 2026 returned five live abstracts (Kiss integral CBF; Molnár input-delay ISS-safety; Capelli; a routing-cost game; interconnected delay functionals). All miss the no-optimizer plus observation-tube conjunction.

2.4 Tubes, liveness modes, and learned shields

Koulong and Pakniyat tighten exponential barriers with robust positively invariant disturbance tubes inside distributed MPC [18]. Those tubes are tracking-error RPI sets for a nominal planner, not observation-age tubes inside a closed-form reflex. Social mini-games add a discrete-time CBF liveness mode and still solve a

[1] Matrix sha256 prefix `83641466...`, snapshot `2026-08-28T16:45:22Z`. Delay-CBF query set re-run the same day; five live hits, none occupying.

Table 1: Conservative miss list against the charter conjunction. Not a novelty warrant. Live snapshot `83641466...`.

Method	Has	Miss
ORCA/RVO2 [1]	Reciprocal VO, no communication	Delay, bounded error, acceleration, sampled-data certificate
HOCBF-QP [5, 7]	Pairwise disc barrier	Online QP; naive wrap optimistic under delay
B-UAVC [2]	Buffered Voronoi	Gaussian chance $\neq$ adversarial tubes; no delay certificate
SMG / MGR [19, 20]	Liveness mode or roundabout	Online QP; delay not certified on this plant
CBFals / ICBF [8, 11]	State or input delay	QP; delay is dynamics/actuation, not observation wrap
Capelli [12]	Connectivity CBF + delay heuristic	Communication; heuristic, not a tube certificate
Ballotta [13]	Distributed CBF + GNN predictor	Communication delay; learned controller
Koulong [18]	RPI-tube tightened eCBF	Disturbance RPI + distributed MPC
ZOCBF / MCBF [14, 15]	Sampled-data barrier	QP or SDP filter
RAMPS [21]	Delay $\wedge$ CBF on the abstract	Learned multi-step shield

QP; on the frozen asynchronous delay corpus the faithful reimplementation is 7/8 [19]. Merry-Go-Round adds a temporary roundabout and still filters through a QP [20]. ICLR 2026 RAMPS states delay and CBF on the abstract and is a learned multi-step shield [21].

3 Plant, observation, and class C

Each robot is a disc of radius 0.25 (pairwise radius-sum R=0.5) with limits $v_{\max}$=2 and $a_{\max}$=2. Control is zero-order held at step h=0.05. The mission plant projects incoming velocity onto the speed ball *before* the position update (`pre_integrate`): if u is the clipped acceleration,

$$\begin{aligned} v^- &= \Pi_{\|v\| \le v_{\max}}(v), \\ p^+ &= p + v^- h + \tfrac{1}{2} u h^2, \\ v^+ &= \Pi_{\|v\| \le v_{\max}}(v^- + uh). \end{aligned} \tag{1}$$

The legacy `post_integrate` convention integrates with the raw incoming velocity and can travel farther than $v_{\max} h$ in one step. It is not the claim plant.

Ballistic identity. Danger is closing speed, not range. Over an unclipped arc of duration h, endpoint clearance against radius-sum is $s - v_{\text{close}} h - R$ (identity max absolute error 8.88×10^{-16} on 11,907 plant-legal cells; `5fac2155...`). Isotropic half-width 54 is an unclipped-arc counterexample (s=3, v_{close}=54, clearance -0.2), not a proof task for (1): the same sample does not collide under `pre_integrate` (`819c83f1...`).

Observation. The charter convention reconstructs the neighbor as delayed neighbor state minus current own state. Age, position-error, and velocity-error bounds on class C are

$$\tau \le 0.2, \qquad e_p = 0.1, \qquad e_v = 0.25.$$

There is no communication channel with which to share inputs or clocks.

Class C. Named two-agent head-on / offset crossing is the open box

$$r_x \in [1.6, 4.0], \quad |r_y| \le 0.35, \quad v_{\text{close}} \in [0.5, 2.0],$$

with goals at distance 4 and ranking $\Phi = \frac{1}{2}(d_{\text{own}} + d_{\text{neighbor}})$. The containing AABB E pads a zero-closing face: $v_x \in [-2, 0]$, $v_y \in [-2, 2]$. The coordinate v_x^{hi}=0 is part of the named envelope and is not a defect to "fix." The ranking box used for progress covers excludes that face ($v_x \in [-2, -0.5]$, v_y=0). Because own is pinned at the origin in the relative chart, $\Phi \ge 2$, so the Gate 6 predicate $\Phi \le 0.35$ cannot fire on this chart.

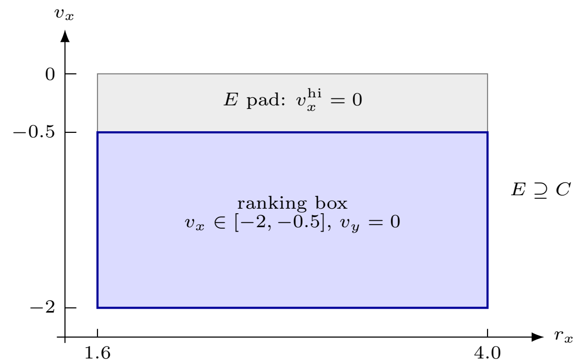

Figure 1: Relative-state (r_x, v_x) slice of the named sets. Envelope E is the containing AABB of class C and includes a zero-closing face $v_x^{\mathrm{hi}} = 0$. Ranking certificates use the stricter closing face $v_x \leq -0.5$. Not to scale in r_y (ranking takes $v_y = 0$; E allows $|v_y| \leq 2$).

4 Closed-form controller

Let r be the observed relative position, $\hat{r} = r/\|r\|$, and $\mathrm{rot}_{90}^{\mathrm{L}}\hat{r}$ the leftward unit tangent. With gains taken from the repaired program `sep_miss_repair_ag2_pe2_ve2_r2_a2.5`,

$$\begin{aligned}
\text{clearance} &= \|r\| - R - \gamma_\tau\, \tau\, v_{\max} - \gamma_p\, e_p, \\
\text{closing} &= -\langle v_{\mathrm{rel}}, \hat{r} \rangle + \gamma_v\, e_v, \\
\alpha &= \max\big(0,\ d_{\mathrm{act}} - \text{clearance}\big), \\
u_{\mathrm{rad}} &= -\gamma_r\, \alpha\, (\text{closing}_+ + 0.25)\, \hat{r}, \\
u_{\mathrm{tan}} &= \gamma_t\, \alpha\, \text{closing}_+\, \mathrm{rot}_{90}^{\mathrm{L}}\hat{r}, \\
u &= \Pi_{\|u\| \leq a_{\max}}\big(u_{\mathrm{goal}} + \textstyle\sum_j (u_{\mathrm{rad},j} + u_{\mathrm{tan},j})\big).
\end{aligned} \tag{2}$$

Here u_{goal} is a PD toward the agent's goal, $\text{closing}_+ = \max(0, \text{closing})$, and the tubes $(\gamma_\tau, \gamma_p, \gamma_v)$ are declared in the clearance, not inferred by a solver. The map is memoryless, left-chiral, and not a Lie-linear CBF constraint, an ORCA halfplane, a Voronoi cell, or a hybrid roundabout (`5c5be098...`).

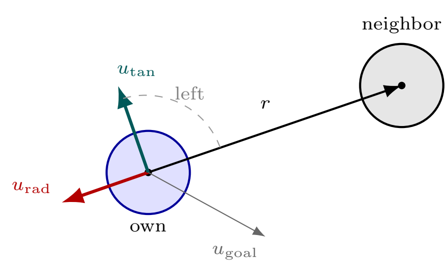

Figure 2: Closed-form pairwise map (2). Radial push is along $-\hat{r}$; the tangent is a fixed left chirality $\mathrm{rot}_{90}^{\mathrm{L}}\hat{r}$. Declared age and error tubes enter only through clearance, not through a solver. Schematic; gains not to scale.

On a plant-legal pairwise grid of n=126 samples the maximum residual against a faithful HOCBF-QP is 3.67; against ORCA it is 3.997. Cosine near-parallel counts are 35/126 (HOCBF) and 41/126 (ORCA). The reflex violates HOCBF slack on 102 samples and ORCA lines on 70. Algebraic markers and the numeric field together reject rediscovery under the recorded transformations (`5c5be098...`). Equation (2) is therefore not a renamed baseline.

5 Observation wrap

Lemma 1 (Collinear constant-velocity wrap)**.** *Fix the charter observation: delayed neighbor minus current own. For a collinear constant-velocity head-on pair with true separation $s > 0$, closing speed $v_c \geq 0$, and age $\tau \geq 0$, the naive reconstructed separation satisfies*

$$s_{\text{naive}} = s + \tfrac{1}{2} v_c \, \tau \tag{3}$$

identically. Consequently the naive barrier $h_{\text{naive}} = s_{\text{naive}}^2 - R^2$ is strictly larger than the true barrier $h = s^2 - R^2$ whenever $v_c\tau > 0$. In particular the sign pattern $h < 0 < h_{\text{naive}}$ is possible.

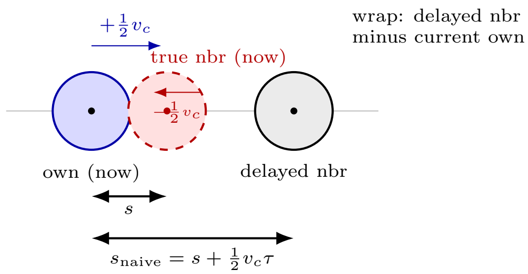

Figure 3: Charter observation wrap, drawn at the tight-crossing contact ($s = 0.487$, $s_{\text{naive}} = 1.305$, age 0.40; `a0986aca...`). True discs overlap ($h < 0$); the naive reconstruction is separated ($h_{\text{naive}} > 1$). Lemma 1 is the collinear constant-velocity identity behind that bias.

Proof. Place own at the origin with velocity $+\frac{1}{2}v_c$ along the line of centers and the neighbor at $+s$ with velocity $-\frac{1}{2}v_c$. After age τ the delayed neighbor position is $s + \frac{1}{2}v_c\tau$, which is exactly the reconstructed relative vector. Expanding $h_{\text{naive}} - h$ gives a nonnegative quantity that is positive for $v_c\tau > 0$. The identity is recorded as `d7a56050...`: on a collinear check $(s, v_c, \tau) = (0.4, 2.0, 0.4)$ the error is 0, with true $h = -0.09$ and naive $h = 0.39$; a 120-cell grid has maximum identity error 0. □

Remark 1. *The default HOCBF wrap in the local reimplementation remains `compensation='none'`. Age-inflation of the neighbor radius by $\tau v_{\max}$ is a diagnostic, not a certified controller: it clears both frozen wrap collisions as a check, not as a replacement for* (2)*.*

Two contacts on the frozen delay corpus instantiate the sign pattern of Lemma 1 on the claim plant (`a0986aca...`):

Case	t	true sep	naive sep	true h	naive h	age
tight crossing	1.15	0.487	1.305	−0.013	+1.45	0.40
close crossing	1.30	0.475	1.169	−0.024	+1.12	0.75

Closing speeds at contact are plant-legal (≈ 1.74 and 0.61; relative max close is 4). This is two contacts, not a dominance theorem.

6 Certificates that stand

All covers in this section use delay-on kinematics, `pre_integrate`, and the repaired program. Envelope E is the containing box of Section 3, including v_x^{hi}=0.

Ranking-box delay-on safety, k=16. Adaptive Lipschitz CEGIS on the named ranking box certifies 3806 leaves, timeout false, minimum certified lower bound 4.92×10^{-5}, depth 16/18 (`60b8d466...`).

Table 2: Controller $\times$ tubes on class C. Standing objects only.

	Closed-form reflex	HOCBF-QP
Declared tubes	Lip. ranking / E-safety stand	infeasible / non-pass-through
No tubes	linearized collision CE	linearized collision CE

Finite-horizon ranking, k**=1...16.** The same box under the ranking predicate (vertex $\Delta\Phi$ plus a Lipschitz remainder strictly negative) certifies k=1, 2, 4, 6 in the ranking-continuum artifact (`a31b0ebc...`) and k=16 with 4268 leaves, remainder-open 0, physical-open 0 (`60b8d466...`). This is a finite-horizon cover from initials in C, not an infinite-horizon ranking theorem and not deadlock-free.

Envelope-E delay-on safety, k**=8.** On E itself the Lipschitz cover certifies k=8 (1132 leaves, physical-open 0; `d7697651...`). The k=12 attempt timed out and was not retried.

Delay-on N-agent half-widths. Compact coupled covers under the class-C envelope and age 0.2 complete at half-widths 0.38 (N=3), 0.52 (N=4), 0.63 (N=5), 0.62 (N=6), 0.45 (N=7), 0.36 (N=8) (`60b8d466...`, `d5d06645...`, `9374a03e...`, `308a1852...`). Not N=41.

Caveat. The Lipschitz family drops its avoidance contribution once $s_{\min}$ exceeds the activation distance passed by the caller (2.0 or 2.5), never inflated by the reflex's own tubes. Avoidance is live to an observed distance of 4.0 and provably off only past true separation 4.5, so all of class C (s_{hi}=4.0) sits in that blind spot. The constant is nevertheless not falsified: an end-to-end probe over the dead zone, the active zone, kink-straddling boxes, four closing speeds, four leaf radii, and horizons k=1, 2, 4, 8, 16 never exceeds budget. Worst measured/claimed ratio is 0.9965 at k=1; a further 129,600-pair position probe reaches 0.99654 and no more (`a7199d84...`). The T^2 over-growth absorbs the missing term. A tube-aware gate would inflate the position Lipschitz factor by up to 8.08$\times$ and cost about 65$\times$ more leaves (4268 $\to$ $\sim$278,000 at k=16). Repair is rejected: it would break covers the probe finds correct. This is not a proof of soundness. It is also not the Taylor status of the next paragraph.

Why the Taylor / ZOH-ball continua are absent. A separate one-shot command-curvature pad used to enclose k-step images was unsound in four independent ways (cutoff past activation, understated $2/s^2$ shape, non-quadratic kinks, and inability to represent `pre_integrate` speed projection; `403f9d55...`). Those defects are why the ZOH-ball and coupled-star continua are withdrawn, not theorems of this paper. Vertex rollouts in the withdrawn artifacts stand as vertices.

7 Necessity matrix

Four hashed cells, one linearized bar (`quadratic=False`, because HOCBF-QP is not C^2), class-C delay-on, `pre_integrate`. Tube-reflex safety on the naive collision box is a ranking-box corollary and is not recertified.

Wrap-none HOCBF. On the compact vertex-collision seed

$$r_x \in [1.60, 1.62], \quad |r_y| \le 0.02, \quad v_x \in [-0.55, -0.50],$$

a center-Jacobian mean-value upper on $\min_t$ separation is $0.373 < 0.5$; all eight vertices collide (min sep 0.464; `a6cbe376...`). A grown box covering the seed's ranking-range r_x and $|r_y| \le 0.14$ remains a continuum CE (worst upper $0.499998 < 0.5$, 1070 leaves; `d245f4cd...`). The vertex neighborhood $|r_y| \le 0.1538$ is not a CE.

HOCBF with error tubes. Transplanting the repaired tubes into the QP does not produce a wrap-none collision CE on that seed (0/8 vertices collide). On a named slow slab of $C^\star$ (volume ratio 0.182 versus $C^\star$) the linearized error-tube flow is a continuum non-pass-through: final $r_{x,\mathrm{lo}} = 1.507 > 0$, vertices fail pass-through, none collide. On a named close-fast box the error-tube QP is continuum-infeasible at k=0 (interval slack lower $1.326 > 0$, all vertices infeasible; `991f6664...`).

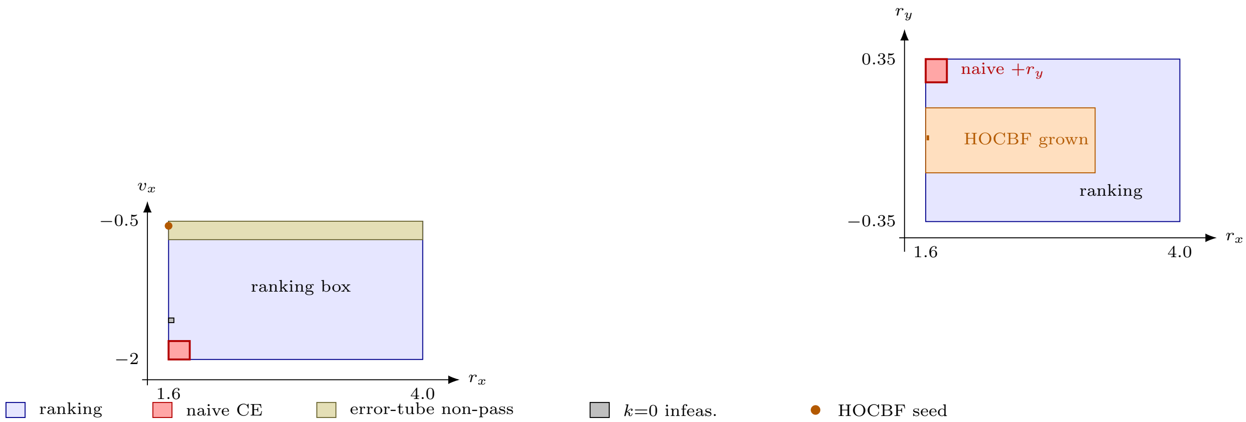

Figure 4: Where the four hashed cells live on class C. Left: (r_x, v_x) inside the ranking box. Tube-less reflex CE (red, $0.475 < 0.5$) is close-fast $+r_y$; wrap-none HOCBF seed (dot, $0.373 < 0.5$) is close-slow; error-tube HOCBF is a slow non-pass-through slab and a k=0 infeasible box, not a collision CE. Right: (r_x, r_y) showing that the naive CE is an offset slab, not the HOCBF seed. Boxes are axis-aligned; the HOCBF seed is a marker because its r_x width is 0.02.

Tube-less seed reflex. The seed program sets age, position, and velocity gains to zero (equivariant_reflex_rg1_tg1.5_ad2_left). That map has a linearized continuum collision CE on the named $+r_y$ close-fast box

$$r_x \in [1.6, 1.8], \quad r_y \in [0.25, 0.35], \quad v_x \in [-2.0, -1.8]$$

(volume ratio 0.00159 versus ranking). Unsplit min-sep upper is $0.475 < 0.5$; 8/8 vertices collide (worst 0.404; `5f6a62da...`). HOCBF's seed is not this seed (0/8 collide). Ranking vertices collide only at $+r_y$ (4/8); the ranking $+r_y$ slab is not a CE (upper 1.808). Quadratic remainder on the seed is an honest miss (Lipschitz branch 72/80, remainder exploded).

Vertex contrast on the naive CE box. Under the class-C envelope, age-aware collides 4/8 (worst 0.460); error-tube collides 0/8 (worst 0.512); repaired collides 0/8 (worst 0.942; `5f6a62da...`). F-001 is a different, harsher envelope and is not recertified.

Tubes are therefore the causal ingredient: stripping them from the closed form produces a collision CE, and giving them to HOCBF produces infeasibility or non-pass-through rather than a live delay-on controller. Only (2) stays live under the tubes. This is not domination of HOCBF on class C.

8 Empirical 54-grid

A 40-step delay-on / `pre_integrate` grid of 54 class-C initials is a table, not a cover (`3abe0935...`, `d3aeb917...`). The repaired reflex is 54/54 collision-free and 54/54 pass-through. Wrap-none HOCBF collides on 17/54. Age-inflate and delay-kinematics wraps collide on 8/54 and 17/54. Error-tube HOCBF collides on 0/54, passes through on 18/54, and freezes on 3/54. Social mini-game collides on 10/54. These counts are empirical. They do not promote the 54-grid to a certificate and they do not say HOCBF is dominated.

9 Limitations

Certificates are finite-horizon and named-set. Class C is a transit class: interval images and vertices leave E. Lipschitz covers are not a proof of soundness of the constant; they are covers plus a probe that did not falsify it. Gate 6 $\Phi \leq 0.35$ does not fire on the relative chart. Hardware is an offline assumption-monitor dry-run, not a live ROS deployment. The literature pass is a miss list with an incomplete proceedings surface (IROS 2026 PaperCept 403). Withdrawn Taylor continua are not quietly restored.

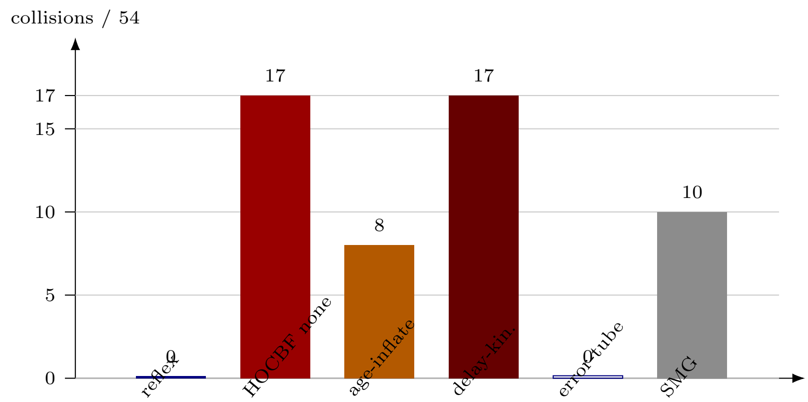

Figure 5: Empirical 54-grid collision counts (delay-on, `pre_integrate`, 40 steps). Reflex and error-tube HOCBF are both 0/54 collisions; error-tube is 18/54 pass-through against the reflex's 54/54. A table, not a cover (`3abe0935...`, `d3aeb917...`).

10 Conclusion

The standing object is small and checkable: a closed-form equivariant reflex with declared observation tubes; a wrap lemma that explains why a naive HOCBF filter is optimistic on this plant; Lipschitz delay-on safety and ranking on named class C, with the gate caveat written next to the cover; and a four-cell matrix whose only live delay-on controller is the closed form with tubes. Gate 7 is unclaimed. The cell may well be occupied tomorrow. Today's hashed artifacts are the paper.

A Standing artifacts

This appendix lists the hashed objects this preprint actually uses. It does not treat the historical fifteen-entry package `crosswind-local-v1` as evidence for the claims above. Content hashes are the `sha256` fields inside each JSON, not raw file hashes.

A.1 Claim objects

- Lipschitz ranking k=1...16 and ranking-box delay-on safety k=16: `a31b0ebc...`, `60b8d466...`
- Envelope-E delay-on safety k=8: `d7697651...`
- Delay-on N-agent covers: N=3 `60b8d466...`; N=4 `d5d06645...`; N=5 `9374a03e...`; N=6, 7, 8 `308a1852...`
- Lipschitz gate audit `C-LIPSCHITZ-GATE-AUDIT`: `a7199d84...`
- Observation wrap lemma: `d7a56050...`; frozen contacts: `a0986aca...`
- Controller non-equivalence: `5c5be098...`
- Wrap-none HOCBF collision CE: `a6cbe376...`, grown `d245f4cd...`
- HOCBF error-tube infeasible / non-pass-through: `991f6664...`
- Tube-less reflex collision CE: `5f6a62da...`
- Empirical 54-grid: `3abe0935...`, tube wrap `d3aeb917...`
- Taylor pad defects (withdrawal reason only): `403f9d55...`

- Restricted synthesis: `dc644f3b...`
- Frontier matrix (28 August 2026): `83641466...`; proceedings miss list (20 August 2026, not recrawled): `8c95e4e7...`; Gate 7 addendum: `5e3cf043...`

A.2 Primary reproduction commands

Locked certificate JSON is not overwritten by these commands. Re-running a CEGIS campaign with a wall-clock budget can change adaptive search metadata; the hashes above are the standing pins.

```
PYTHONPATH=src python3 -m cli.crosswind verify-artifacts
PYTHONPATH=src python3 -m cli.crosswind class-c-ranking-continuum
PYTHONPATH=src python3 -m cli.crosswind class-c-hocbf-continuum
PYTHONPATH=src python3 -m cli.crosswind class-c-hocbf-tube-continuum
PYTHONPATH=src python3 -m cli.crosswind class-c-naive-reflex-continuum
PYTHONPATH=src python3 -m cli.crosswind lipschitz-gate-audit
PYTHONPATH=src python3 -m cli.crosswind cbf-observation-mismatch
PYTHONPATH=src python3 -m cli.crosswind cbf-observation-lemma
PYTHONPATH=src python3 -m cli.crosswind controller-equivalence
```

Live arXiv refresh (`scripts/refresh_frontier.py --live-arxiv`) updates the frontier matrix hash and is not required to check the certificate pins. PaperCept is not part of this reproduction.

References

[1] Jur van den Berg, Stephen J. Guy, Ming Lin, and Dinesh Manocha. Reciprocal n-body collision avoidance. In *Robotics Research*, volume 70 of *Springer Tracts in Advanced Robotics*, pages 3–19. Springer, 2011.

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[3] Aaron D. Ames, Xiangru Xu, Jessy W. Grizzle, and Paulo Tabuada. Control barrier function based quadratic programs for safety critical systems. *IEEE Transactions on Automatic Control*, 62(8):3861–3876, 2017.

[4] Wei Xiao and Calin Belta. Control barrier functions for systems with high relative degree. In *Proc. IEEE Conf. Decision and Control*, pages 627–634, 2019.

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